ACTL2102 Tutorial

Week 9: Non-Homogeneous Markov Chains and Multiple-State Models

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T3 2026

Plan

- Recap: from homogeneous to non-homogeneous chains, occupancy and the next jump, probabilities of paths
- Main questions: **4.8, 4.9, 4.7, 4.18, 4.16**
- If time permits: 4.6, 4.15

From Week 8: Homogeneous to Non-Homogeneous

	Homogeneous (Weeks 7–8)	Non-homogeneous (Week 9)
Rates	q_{ij}, constant	$q_{ij}(t)$, vary with time
Transition probabilities	$P_{ij}(t)$: elapsed time	$P_{ij}(s,t)$: both times
Holding times	$T_i \sim \text{Exp}(\nu_i)$	not exponential in general
Next state	q_{ij}/ν_i, independent of T_i	$q_{ij}(u)/\nu_i(u)$ for a jump at time u
Occupancy	$e^{-\nu_i t}$	$\exp\left(-\int_s^t \nu_i(u)\,du\right)$
Kolmogorov equations	constant coefficients	time-varying coefficients

The rates change with time, for example with age, so every probability depends on when it starts as well as how long it runs: there are always at least two times to track.

Non-Homogeneous Markov Chains

The transition probabilities depend on both times:

$$P_{ij}(s,t) = \Pr[X(t) = j \mid X(s) = i], \qquad s \le t.$$

The rate $q_{ij}(t)$ is defined by $P_{ij}(t, t+h) = q_{ij}(t)\,h + o(h)$ for $j \neq i$, and $\nu_i(t) = \sum_{k \neq i} q_{ik}(t) = -q_{ii}(t)$ is the rate of leaving i at time t. Together they form $Q(t)$.

The chain is Markov: given $X(s) = i$, its path after s does not depend on its path before s. So the Chapman–Kolmogorov equations hold:

$$P(s,t) = P(s,u)\,P(u,t) \quad \text{for } s \le u \le t, \qquad P(s,s) = I.$$

Kolmogorov Equations

Backward equations, in s:

$$\frac{\partial}{\partial s} P(s,t) = -Q(s)\, P(s,t), \qquad \frac{\partial}{\partial s} P_{ij}(s,t) = -\sum_k q_{ik}(s)\, P_{kj}(s,t).$$

Forward equations, in t:

$$\frac{\partial}{\partial t} P(s,t) = P(s,t)\, Q(t), \qquad \frac{\partial}{\partial t} P_{ij}(s,t) = \sum_k P_{ik}(s,t)\, q_{kj}(t).$$

Both are solved from $P(s,s) = I$. With time-varying coefficients there is no general closed form, and numerical methods are outside the scope of this course.

Exercise 4.8: Alive–dead model

Exercise 4.8: [cmc7, Solution] A life aged x can be either alive or dead. State A is alive, D is dead. The transition rate from state A to state D at age y is $\mu(y)$.

1. Using the Kolmogorov forward equation, derive an expression for ${}_tp_x$, the probability that an individual currently (i.e., at $t = 0$) alive with current age x will be alive at age $x + t$ (at time t).

2. Write down the Chapman-Kolmogorov equations and use them to prove that:

$$ {}_{z-x}p_x = {}_{y-x}p_x \cdot {}_{z-y}p_y. $$

Occupancy Probabilities

As in Week 8, the **occupancy probability** $\overline{P}_{ii}(s,t)$ is the probability of staying in i throughout $[s,t]$, given $X(s) = i$; the lecture writes $P_{\overline{ii}}(s,t)$. The bar matters: the transition probability $P_{ii}(s,t)$ also counts paths that leave i and return.

The probability of staying is still exponential, but the rate changes, so the exponent integrates it:

$$e^{-\nu_i(t-s)} \quad \text{becomes} \quad \overline{P}_{ii}(s,t) = \exp\Big(-\int_s^t \nu_i(u)\,du\Big).$$

In the survival model, with force of mortality $\mu(t)$, death is absorbing, so being alive at t means never having left: $P_{AA}(s,t) = \overline{P}_{AA}(s,t) = \exp\big(-\int_s^t \mu(u)\,du\big)$.

Residual Holding Time and the Next Jump

Given $X(s) = i$, the **residual holding time** $R(s)$ is the random time from s to the next jump, the non-homogeneous version of T_i. It exceeds w exactly when the chain stays in i throughout $[s, s+w]$, so its survival function is an occupancy probability:

$$\Pr(R(s) > w \mid X(s) = i) = \overline{P}_{ii}(s, s+w),$$

and its density is $\nu_i(s+w)\,\overline{P}_{ii}(s, s+w)$. The state entered at that jump, $X^{(+)}(s) = X(s + R(s))$, is j with probability

$$\Pr[X^{(+)}(s) = j \mid X(s) = i,\ R(s) = w] = \frac{q_{ij}(s+w)}{\nu_i(s+w)}.$$

Exercise 4.9: Gompertz mortality and reaching age 90

Exercise 4.9: [cmc8, Solution] A retirement savings account provider provides a death benefit insurance to the account holders. A death benefit is a payment to the dependent of a deceased at the moment of deceasing. The payment of death benefit is once every five years to the account provider. Obviously, the probability of dying depends of the age (x) of the insured and for $x \geq 65$ the rate of dying is given by:

$$\mu(x) = \exp(\alpha + \beta \cdot x),$$

where for the Australian population the parameters are $\alpha = -11$ and $\beta = 0.1$.

1. Define the transition intensity matrix $\mathbf{A}(x)$.
2. Give an expression for the yearly transition probabilities for an x year old individual.
3. Calculate the probability of reaching age 90 for a 65 year old Australian.
4. Give the distribution of the residual holding time.

Exercise 4.7: Duration-dependent rates

Exercise 4.7: [cmc6, Solution] Throughout a person's working life he or she may be Employed (state E) or Unemployed (state U). At age t the individual is in state X_t and has been in that state since age $t - V_t$. The probability of finding employment between age t and age $t + h$, given that $X_t = U$ and $V_t = v$, is $\sigma_{UE}(t; v) \cdot h + o(h)$. The corresponding probability of becoming unemployed between age t and $t + h$, given that $X_t = E$ and $V_t = v$, is $\sigma_{EU}(t; v) \cdot h + o(h)$.
Note: In contrast to the usual case, in this case the transition rates can depend on the time spend in the state $t - v$.

1. Does the process $X = \{X_t : t \geq 0\}$ satisfy the Markov property?
   Would your answer change if σ_{UE} and σ_{EU}, were:
   (a) independent of v,
   (b) independent of t?

2. If $\sigma_{UE}(t; v) = \sigma_{EU}(t; v) = \lambda$ per year for all t and all v, write down the distribution of the number of transitions which take place in the interval for age 18 to age t years, and hence find an expression for $\Pr(X_t = E | X_{18} = E)$.

3. The individual begins a period of unemployment at age t_0. If $S(v)$ denotes the probability that the individual remains unemployed throughout the age interval $(t_0, t_0 + v)$ show that:

$$\frac{d}{dv} S(v) = -\sigma_{UE}(t_0 + v, v) \cdot S(v).$$

   Hint: use: $\frac{d}{dx} f(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$.

4. Evaluate $S(v)$ in the case where $\sigma_{UE}(t; v) = \alpha \cdot \beta^{-\alpha} \cdot v^{\alpha - 1}$.

5. Assuming that the parameter values are given by $\alpha = 0.35$ and $\beta = 0.1$, calculate the probability that the period of unemployment lasts for:
   (a) at least one year;
   (b) at least two years.

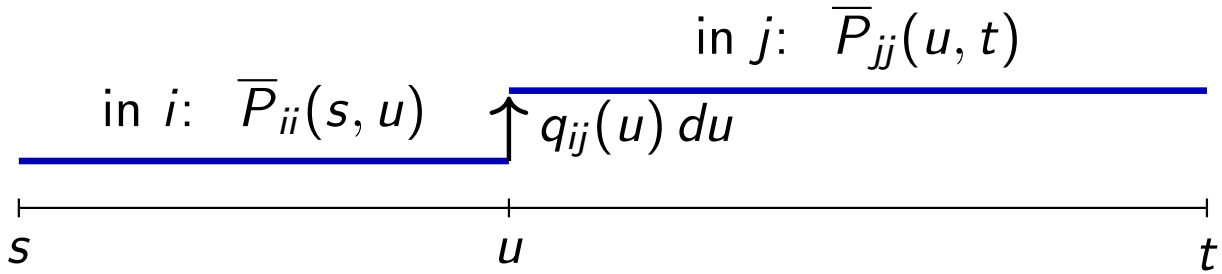
Probabilities of Paths

Given $X(s) = i$, the path that stays in i until u, jumps to j in $[u, u + du)$ and stays in j until t has probability

$$\overline{P}_{ii}(s, u)\; q_{ij}(u)\, du\; \overline{P}_{jj}(u, t),$$

by the Markov property at the jump. This is Week 8's first-jump integrand, with each rate taken at its own time. When the jump can happen anywhere in a window $[a, b]$, integrate over $u \in [a, b]$.

Conditioning on the time of the first or the last jump in this way gives the **integrated Kolmogorov equations**, backward and forward.



Exercise 4.18: Disability insurance

Exercise 4.18: [cmc17, Solution] An insurer provides a disability insurance, where there will only be a payment if the insured is disabled for more then 2 years. The transition intensities are:

- The rate of dying for a healthy individual is $0.001 + (x - 25)/4000$;
- The rate of dying for a disabled individual is $0.02 + 3 \cdot (x - 25)/4000$;
- The rate of becoming disabled is $\log(x - 23)/10$;
- The rate of recovering from disability is $(x - 23)^{-0.5}$.

Hint: $\int_a^b \log(x)dx = [x\log(x) - x]_a^b$.

1. Determine the probability of staying healthy for an insured aged x for the next t years.
2. Determine the probability of staying disabled for an insured aged x for the next t years.
3. Describe, in words, the effect of age on the probability of recovering and the probability of becoming disabled.
4. Give an expression for the probability for a 40 year old insured who is currently disabled to remain disabled in for at least 2 years and then recover and remain healthy till the end of the third year.
   Note: give your answer as $\int_a^b f(t)dt$, do not try to solve the integral.
5. Give an expression for the probability for a 40 year old insured currently disabled to remain disabled and then die in the second year.
   Note: give your answer as $\int_a^b f(t)dt$, do not try to solve the integral.

Exercise 4.16: At most w years sick

Exercise 4.16: [cmc15, Solution] Consider the following disability insurance model with three states: Healthy (H), Sick (S) and Dead (D). The transition intensity from H to S is, σ_x, from H and D is μ_x, from S to H is $\rho_{x,z}$ and from S to D is $\nu_{x,z}$. For those intensities, x denotes the current age of the life and z denotes the duration of the current time spent in state S. Let:

$$ {}_tP_x^{\overline{HH}} = \Pr\left(\text{state remains in H until age } x+t \,\middle|\, \text{state at age } x \text{ is } H\right) $$

and

$$ {}_{w,t}P_x^{\overline{HS}} = \Pr\left(\begin{array}{l} \text{state at age } x+t \text{ is } S \text{ with duration of current time spent in state } S \leq w, \\ \text{and there is no transitions before entering the state S} \end{array} \middle| \text{state at age } x \text{ is } H \right). $$

1. Write down a formula for ${}_{w,t}P_x^{\overline{HS}}$ in terms of ${}_uP_y^{\overline{HH}}$ and the transition intensities.
2. Now suppose that for all values of x and z:

$$ \begin{aligned} \sigma_x &= 0.05, \\ \mu_x &= 0.005, \\ \nu_{x,z} &= 0.01, \\ \rho_{x,z} &= 0. \end{aligned} $$

Calculate the value of ${}_{2,3}P_{30}^{\overline{HS}}$.

Notation from This Week

$P_{ij}(s,t)$	probability of being in j at time t, given i at time s, by any route
$q_{ij}(t)$, $\nu_i(t)$	rate of moves $i \to j$ at time t; total rate of leaving i at time t
$\overline{P}_{ii}(s,t)$	occupancy: stays in i throughout $[s,t]$; equals $\exp\left(-\int_s^t \nu_i(u)\,du\right)$
$\mu(t)$	force of mortality: the rate from alive to dead
$R(s)$	residual holding time: from s to the next jump; survival function $\overline{P}_{ii}(s, s+w)$
$X^{(+)}(s)$	state entered at that jump
One jump at u	$\overline{P}_{ii}(s,u)\,q_{ij}(u)\,du\,\overline{P}_{jj}(u,t)$, integrated over the window for u

Exercise 4.6: Non-homogeneous Poisson process

Exercise 4.6: [cmc5, Solution] **Non-Homogeneous Poisson Process**: The total number of claims received by an insurance company is described by a non-homogeneous Poisson process with rate $\lambda(t)$.

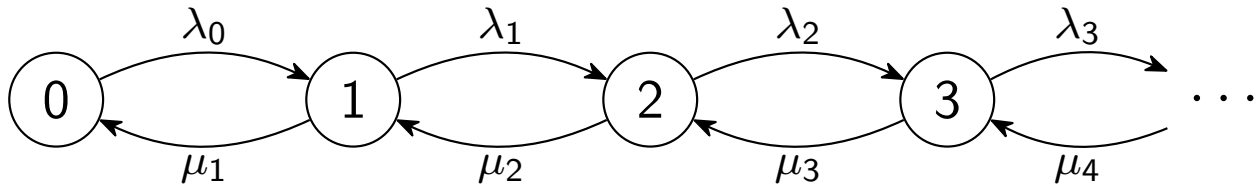
1. Write down the Kolmogorov forward equations for this non-homogeneous process.
2. Show that the solution is of the form:

$$P_{0j}(s,t) = \frac{(m(s,t))^j \cdot e^{-m(s,t)}}{j!}$$

and specify the form for $m(s,t)$.

Birth and Death Processes

A **birth and death process** is a homogeneous chain on $\{0, 1, 2, \dots\}$ that moves one step at a time: from n to $n+1$ at rate λ_n (a birth) and from n to $n-1$ at rate μ_n (a death), with $\mu_0 = 0$. So $\nu_n = \lambda_n + \mu_n$, and Q is tridiagonal.



In the long run, the balance equations of Week 8 reduce to

$$\lambda_n P_n = \mu_{n+1} P_{n+1}, \qquad n = 0, 1, 2, \dots :$$

the rate of moves up from n equals the rate of moves down from $n+1$.

Exercise 4.15: Population with restricted immigration

Exercise 4.15: [cmc14, Solution] Ross: Chapter 6, Q12. Each individual in a biological population is assumed to give birth at an exponential rate λ, and to die at an exponential rate μ. In addition, there is an exponential rate of increase θ due to immigration. However, immigration is not allowed when the population size is N or larger.

1. Set this up as a birth and death model.
2. If $N = 3$, $1 = \theta = \lambda$, and $\mu = 2$, determine the proportion of time that immigration is restricted. **Hint**: You might use: $\sum_{k=1}^{\infty} \frac{1}{k2^k} = \log(2)$.

Thanks for coming!